Generalist Model for Structured Data: Foundations, Frontiers and Applications

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Abstract

Structured data constitutes one of the most ubiquitous data modalities in web-scale and enterprise applications, supporting tasks such as recommendation, forecasting, and user behavior analysis. Conventional modeling paradigms—ranging from generalized linear models and gradient boosting to deep structured networks—have provided strong baselines for predictive analytics and decision support. However, the recent emergence of foundation models and in-context learning (ICL) has sparked a new paradigm for structured modeling, moving from dataset-specific training toward universal, adaptable inference. Emerging structured foundation models illustrate how large-scale pretraining, synthetic data generation, and ICL-based inference can extend foundation-model principles to structured data. These developments open new possibilities for multi-task learning, zero-shot inference, and knowledge transfer across diverse structured settings. Yet, the space of structured foundation models remains largely unexplored, with open questions surrounding data generation, multi-task setting, pretraining objectives, and evaluation standards. This tutorial will provide a structured overview of both conventional modeling and recent ICL-based approaches. Participants will gain a comprehensive understanding of established methods, current advances in foundation models, and open research challenges. In particular, we will offer an in-depth introduction to structured ICL and review the most representative foundation models in this field. Several key topics in this field will be discussed, including pretraining data generation, multi-task learning, and other emerging directions in the modeling of structured data. This tutorial aims to bridge conventional machine learning and the emerging foundation-model paradigm, providing attendees with conceptual and practical insights into structured data modeling in the era of generalist foundation models.

CCS Concepts

• **Computing methodologies → Artificial intelligence**.

Keywords

Structured Data, Foundation Model, In-Context Learning

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1 Topic and Relevance

In real-world applications, massive amounts of structured data are continuously generated, forming the backbone of numerous domains. Typical forms of structured data include tabular data, time-series and graph, which are pervasive across web systems and online platforms; among these modalities, tabular and time-series data constitute the primary focus, garnering substantial interest from both the industrial and academic communities. To ensure the efficient operation of online systems, it is essential to accurately model the underlying relationships and causal structures within structured data [2, 31]. Conventionally, modeling structured data involves classical statistical and machine learning approaches. For tabular data, the methods include linear regression, decision trees, random forests, and gradient boosting machines (e.g., XGBoost [7], CatBoost [22]) that leverage the explicit feature structure to learn patterns and make predictions, they have dominated this domain for a decade due to their strong performance, native handling of heterogeneous features, and robustness in relatively small- to medium-scale datasets [9, 10, 25]. For time-series and graph data, the developments follow a similar trajectory. Traditional time-series modeling approaches, such as ARIMA [3] and ETS [16], have been

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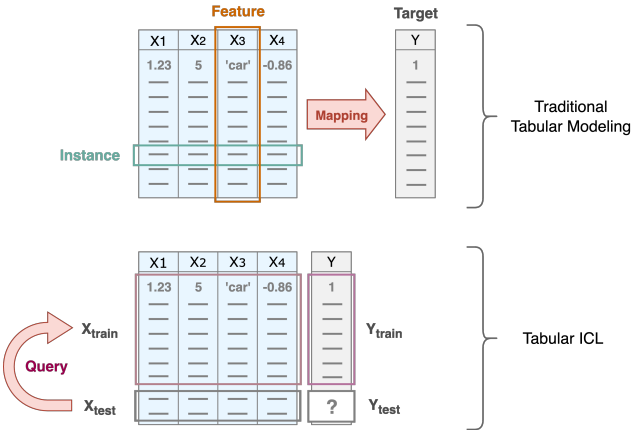

Figure 1: Traditional structured data modeling vs. structured data ICL.

outperformed by deep learning methods, which have inspired a substantial body of subsequent research [11–13, 19]. Likewise, classical probabilistic and stochastic graph models have largely been replaced by data-driven approaches [30]. In the domain of structured data, conventional approaches exhibit several notable limitations. They often assume feature independence and require extensive preprocessing, including categorical encoding, handling of missing values, and feature scaling to achieve satisfactory performance. Moreover, these methods suffer from well-recognized drawbacks such as heavy dependence on dataset-specific feature engineering and hyperparameter tuning, limited transferability across tasks or domains, and inefficiency in dynamic, heterogeneous, web-scale environments that demand rapid adaptation. As noted in [6, 17, 27], the field is still struggling to move beyond per-dataset modeling.

In recent years, the emergence of foundation-model-style and ICL paradigms has offered a significant leap forward for structured data modeling [8, 14, 33]. The key idea is that a model can learn to perform a new task at inference time by conditioning on a set of labeled examples (the "context"), without gradient-based parameter updates. Applied to structured data, ICL allows a model to make predictions on a new table by leveraging context rows from that table or from related tables, rather than retraining on each new dataset. Using tabular data as an example, as shown in Figure 1, assuming a relationship exists between the features and the target (typically a classification label or regression variable), traditional structured data modeling aims to train or fine-tune a model to learn a robust mapping from X to Y. However, such an approach struggles to handle domain shifts, as the learned mapping may be dataset-specific and fail to generalize across different tables. In structured data ICL, the underlying logic shifts: instead of learning a single, dataset-specific mapping, the model is trained to determine, for each instance in the test set X_{test}, how to identify similar instances from the context—i.e., the knowledge base X_{train}. Based on this retrieved context, the unknown targets in Y_{test} can be inferred through a non-linear mapping from the corresponding values in Y_{train}, enabling flexible and adaptive predictions across different datasets. From this perspective, ICL emerges as a promising avenue for developing highly generalizable and adaptable solutions, garnering increasing attention from both the research community and industry. For instance, [14] introduced TabPFN-v1, a model capable of zero- and few-shot learning on new structured data tasks without dataset-specific retraining, marking the first public demonstration of tabular foundation models. Subsequently, the authors released a major update, TabPFN-v2, which offers improved performance and reliability [15]. Similarly, TabICL [23] extends ICL to large-scale tables with architectures designed for efficient context processing. More recently, LimiX [32] has unified classification, regression, and missing value imputation within a single model, garnering considerable attention in the field and highlighting the potential of unified foundation models. These developments indicate a paradigm shift—from bespoke, per-task modeling toward unified modeling of structured data capable of rapid adaptation, transfer-learning across schemas and domains, and minimal task-specific tuning.

Although ICL has rapidly advanced, several critical open problems remain before it can reliably replace conventional methods: first, how to design synthetic pretraining data that captures real-world variable distributions, constraints, categorical imbalances, and causal structure so pretrained models transfer robustly to diverse downstream tables — current work shows synthetic pretraining can enable powerful ICL but also highlights gaps in realism and constraint handling [15, 21, 24]. Second, multi-task and multi-modal learning for ICL requires better task-conditioning schemes and unified objectives so a single backbone can generalize across very different prediction targets and missingness patterns [32]. Third, efficient scaling and context management — i.e., how to represent, retrieve, or compress very large support sets without losing salient examples — remains a bottleneck for applying ICL to large datasets and streaming settings [26]. Finally, fine-tuning and adaptation strategies that safely and effectively combine a frozen pretrained ICL learner with small amounts of task data (or that tune only adapters / retrieval components) are underexplored; we need clear protocols and benchmarks to measure when fine-tuning improves performance versus when it undermines the in-context learning algorithm. This introduces critical challenges for the current evaluation frameworks [1, 20].

For the web-scale community, this trend is particularly timely: online services generate massive structured logs, heterogeneous database tables, and evolving schemas, making the ability to generalise across datasets, adapt quickly, and minimise manual engineering especially valuable. A tutorial that walks through both the established practice of structured data modeling and the emerging frontier of the foundation models of structured data thus fills a vital gap: it enables practitioners and researchers in web-systems, analytics, and AI-driven services to understand how structured data modeling is changing and why these changes matter. By covering the fundamentals and then exploring the ICL/foundation-model frontier.

2 The Tutorial Team

The tutorial will be presented by researchers from both academia and industry, who bring extensive experience in foundation models and structured data modeling. Together with their collaborators, they have published in top-tier venues such as KDD, WWW, ICLR,

AAAI, IJCAI, etc; covering topics across graph modeling [28, 29], time series analysis [13, 18] and tabular modeling [4, 5, 31, 32]. In addition, all presenters have prior teaching and presentation experience at their respective institutions and major international conferences.

- **Peng Cui** is an Associate Professor with tenure in the Department of Computer Science at Tsinghua University. He got his PhD degree from Tsinghua University in 2010. His research interests include causally-regularized machine learning, network representation learning, and social dynamics modeling. He has published more than 100 papers in prestigious conferences and journals in data mining and multimedia. His recent research won the IEEE Multimedia Best Department Paper Award, SIGKDD 2016 Best Paper Finalist, ICDM 2015 Best Student Paper Award, SIGKDD 2014 Best Paper Finalist, IEEE ICME 2014 Best Paper Award, ACM MM12 Grand Challenge Multimodal Award, and MMM13 Best Paper Award. He is PC co-chair of CIKM2019 and MMM2020, SPC or area chair of ICML, KDD, WWW, IJCAI, AAAI, etc., and Associate Editors of IEEE TKDE, IEEE TBD, ACM TIST, and ACM TOMM etc. He received ACM China Rising Star Award in 2015, and CCF-IEEE CS Young Scientist Award in 2018. He is now a Distinguished Member of ACM and CCF, and a Senior Member of IEEE.
- **Xingxuan Zhang** is an Assistant Researcher in the Department of Computer Science at Tsinghua University, where he also received his PhD. His research interests lie in Foundation Models and Trustworthy AI, with a specialized focus on developing fundamental modeling methods for structured data. He has published more than 30 papers in top-tier conferences and journals including ICLR, Neurips, ICML, WWW, CVPR, and ICCV, where he also served as a reviewer and program committee member.
- **Han-Jia Ye** is an Associate Professor in the School of Artificial Intelligence at Nanjing University. His research centers on machine learning—especially representation learning, meta-learning, model reuse, and deep learning for tabular data. He received his Ph.D. in Computer Science from Nanjing University in 2019. He has served as Tutorial Co-Chair for SDM 2023 and Doctoral Forum Co-Chair for SDM 2022, and as an area chair for top-tier venues including ICML, NeurIPS, ICLR, AAAI, IJCAI, and CVPR. He leads the development of TALENT, a toolbox of representative deep tabular methods, and is organizing an AAAI 2026 tutorial on representation learning for structured tabular data.
- **Jintai Chen** is an Assistant Professor in the AI Thrust, Information Hub at The Hong Kong University of Science and Technology (Guangzhou). He received his PhD from Zhejiang University and later conducted postdoctoral research at the University of Illinois at Urbana–Champaign. His research centers on artificial intelligence for tabular data and healthcare, with a particular focus on developing foundation models for heterogeneous and noisy clinical and biomedical datasets. His broader interests include medical agents, clinical trial optimisation, and drug design.
- **Shuyang Li** is a Postdoctoral Researcher in the Department of Computer Science at Tsinghua University. He received his PhD degree in Electrical, Electronic, and Communications Engineering from Politecnico di Torino, Italy. After obtaining his PhD, he has been working as a Postdoctoral Researcher at Politecnico di Torino, collaborating on projects with Telecom Italia, and conducting research in mobile traffic forecasting and predictive maintenance. His research has been published in journals and conferences such as Computer Networks, ICC, WCNC, VTC, etc. His main research interests include time series analysis, synthetic data generation, cellular network optimization, and mobile traffic modeling.

3 Style

This tutorial will be delivered in a lecture-style format and aims to provide a comprehensive introduction to research on the foundation models of structured data, covering pioneering work, state-of-the-art methods, and emerging challenges and future research directions.

4 Schedule

The detailed tutorial schedule is as follows:

Section 1. Introduction
Duration: 30 mins
Presenter: Peng Cui
Content:
- Introduction of structured data modeling
- The bottlenecks of traditional methods
- From task-specific to task-agnostic Learning

Section 2. Conventional modeling of structured data
Duration: 40 mins
Presenter: Han-Jia Ye and Shuyang Li
Content:
- Statistical and probabilistic foundations
- Tree-based ensembles
- Deep learning for structured data
- Empirical performance and limitations

Section 3. ICL of structured data
Duration: 40 mins
Presenter: Xingxuan Zhang and Shuyang Li
Content:
- Introduction of ICL and foundation models
- General design of foundation models
- Pretraining objectives and strategies
- Fine-tuning and retrieval

Section 4. Analysis of advanced methods
Duration: 50 mins
Presenter: Jintai Chen and Xingxuan Zhang
Content:
- Introduction of SOTA approaches
- Comparison of the models

Section 5. Open problems and future directions
Duration: 20 mins
Presenter: Peng Cui
Content:
- Where we stand and where we're headed
- Emerging research directions

5 Audience and Prerequisite Knowledge

This tutorial is designed for participants at all levels and aims to engage a diverse audience from both academia and industry. No specific background knowledge is required. The tutorial offers a systematic overview of the structured data learning domain, beginning with the fundamentals of structured data modeling and ICL, and progressing to advanced topics such as the design of foundation models, pretraining data generation, retrieval mechanisms, and related developments. To enhance engagement, a basic understanding of machine learning and deep learning is recommended; however, the tutorial will also cover the necessary foundational concepts to ensure accessibility for all attendees.

6 Tutorial Materials

We provide extensive materials to help attendees gain a deeper understanding of the topics, including the following:

- **Video Teaser**: The video teaser is available at: https://youtu.be/ZYChOyxs4-Y.
- **Slides**: The slides in our presentation will be released publicly.
- **Related Papers**: A reading list of selected papers will be provided, and we confirm that all included materials are free from copyright restrictions.

In this tutorial, each section will feature a Q&A session, and attendees are encouraged to ask questions at any time during the tutorial.

References

[1] Bernd Bischl, Bernd Bischl, Giuseppe Casalicchio, Matthias Feurer, Pieter Gijsbers, Frank Hutter, Michel Lang, Rafael Gomes Mantovani, Jan van Rijn, and Joaquin Vanschoren. 2021. OpenML Benchmarking Suites. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, J. Vanschoren and S. Yeung (Eds.), Vol. 1.

[2] Vadim Borisov, Tobias Leemann, Kathrin Seßler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci. 2022. Deep neural networks and tabular data: A survey. *IEEE transactions on neural networks and learning systems* 35, 6 (2022), 7499–7519.

[3] George EP Box, Gwilym M Jenkins, Gregory C Reinsel, and Greta M Ljung. 2015. *Time series analysis: forecasting and control*. John Wiley & Sons.

[4] Jintai Chen, KuanLun Liao, Yanwen Fang, Danny Chen, and Jian Wu. 2023. [From Tutor Team] Tabcaps: A capsule neural network for tabular data classification with bow routing. In *The Eleventh International Conference on Learning Representations*.

[5] Jintai Chen, Kuanlun Liao, Yao Wan, Danny Z Chen, and Jian Wu. 2022. [From Tutor Team] Danets: Deep abstract networks for tabular data classification and regression. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 3930–3938.

[6] Jintai Chen, Jiahuan Yan, Qiyuan Chen, Danny Z Chen, Jian Wu, and Jimeng Sun. 2024. [From Tutor Team] Can a deep learning model be a sure bet for tabular prediction?. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 288–296.

[7] Tianqi Chen and Carlos Guestrin. 2016. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*. 785–794.

[8] Xi Fang, Weijie Xu, Fiona Anting Tan, Jiani Zhang, Ziqing Hu, Yanjun Qi, Scott Nickleach, Diego Socolinsky, Srinivasan Sengamedu, and Christos Faloutsos. 2024. Large Language Models (LLMs) on Tabular Data: Prediction, Generation, and Understanding–A Survey. *arXiv preprint arXiv:2402.17944* (2024).

[9] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. 2021. Revisiting deep learning models for tabular data. *Advances in neural information processing systems* 34 (2021), 18932–18943.

[10] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. 2022. Why do tree-based models still outperform deep learning on typical tabular data? *Advances in neural information processing systems* 35 (2022), 507–520.

[11] Lu Han, Xu-Yang Chen, Han-Jia Ye, and De-Chuan Zhan. 2024. [From Tutor Team] Softs: Efficient multivariate time series forecasting with series-core fusion. *Advances in Neural Information Processing Systems* 37 (2024), 64145–64175.

[12] Lu Han, Han-Jia Ye, and De-Chuan Zhan. 2024. [From Tutor Team] SIN: Selective and interpretable normalization for long-term time series forecasting. In *Forty-first International Conference on Machine Learning*.

[13] Lu Han, Han-Jia Ye, and De-Chuan Zhan. 2024. [From Tutor Team] The capacity and robustness trade-off: Revisiting the channel independent strategy for multivariate time series forecasting. *IEEE Transactions on Knowledge and Data Engineering* 36, 11 (2024), 7129–7142.

[14] Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. [n. d.]. TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second. In *NeurIPS 2022 First Table Representation Workshop*.

[15] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. 2025. Accurate predictions on small data with a tabular foundation model. *Nature* 637, 8045 (2025), 319–326.

[16] Rob Hyndman, Anne Koehler, Keith Ord, and Ralph Snyder. 2008. *Forecasting with exponential smoothing: the state space approach*. Springer.

[17] Jun-Peng Jiang, Si-Yang Liu, Hao-Run Cai, Qile Zhou, and Han-Jia Ye. 2025. [From Tutor Team] Representation learning for tabular data: A comprehensive survey. *arXiv preprint arXiv:2504.16109* (2025).

[18] Shuyang Li, Enrico Magli, and Gianluca Francini. 2023. [From Tutor Team] To be conservative or to be aggressive? A risk-adaptive mixture of experts for mobile traffic forecasting. In *ICC 2023-IEEE International Conference on Communications*. IEEE, 5471–5476.

[19] Bryan Lim and Stefan Zohren. 2021. Time-series forecasting with deep learning: a survey. *Philosophical Transactions of the Royal Society A* 379, 2194 (2021), 20200209.

[20] Duncan McElfresh, Sujay Khandagale, Jonathan Valverde, Vishak Prasad C, Ganesh Ramakrishnan, Micah Goldblum, and Colin White. 2023. When Do Neural Nets Outperform Boosted Trees on Tabular Data?. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., 76336–76369. https://proceedings.neurips.cc/paper_files/paper/2023/file/f06d5ebd4ff40b40dd97e30cee632123-Paper-Datasets_and_Benchmarks.pdf

[21] Weronika Ormaniec, Scott Sussex, Lars Lorch, Bernhard Schölkopf, and Andreas Krause. 2025. Standardizing Structural Causal Models. In *The Thirteenth International Conference on Learning Representations*.

[22] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. 2018. CatBoost: unbiased boosting with categorical features. *Advances in neural information processing systems* 31 (2018).

[23] Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. 2025. TabICL: A Tabular Foundation Model for In-Context Learning on Large Data. In *ICML 2025-Forty-Second International Conference on Machine Learning*.

[24] Alexander Reisach, Myriam Tami, Christof Seiler, Antoine Chambaz, and Sebastian Weichwald. 2023. A scale-invariant sorting criterion to find a causal order in additive noise models. *Advances in Neural Information Processing Systems* 36 (2023), 785–807.

[25] Ravid Shwartz-Ziv and Amitai Armon. 2022. Tabular data: Deep learning is not all you need. *Information Fusion* 81 (2022), 84–90.

[26] Valentin Thomas, Junwei Ma, Rasa Hosseinzadeh, Keyvan Golestan, Guangwei Yu, Maksims Volkovs, and Anthony Caterini. 2024. Retrieval & fine-tuning for in-context tabular models. *Advances in Neural Information Processing Systems* 37 (2024), 108439–108467.

[27] Boris Van Breugel and Mihaela Van Der Schaar. 2024. Position: why tabular foundation models should be a research priority. In *Proceedings of the 41st International Conference on Machine Learning*. 48976–48993.

[28] Xiao Wang, Meiqi Zhu, Deyu Bo, Peng Cui, Chuan Shi, and Jian Pei. 2020. [From Tutor Team] Am-gcn: Adaptive multi-channel graph convolutional networks. In *Proceedings of the 26th ACM SIGKDD International conference on knowledge discovery & data mining*. 1243–1253.

[29] Lingfei Wu, Peng Cui, Jian Pei, Liang Zhao, and Xiaojie Guo. 2022. [From Tutor Team] Graph neural networks: foundation, frontiers and applications. In *Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining*. 4840–4841.

[30] Feng Xia, Ke Sun, Shuo Yu, Abdul Aziz, Liangtian Wan, Shirui Pan, and Huan Liu. 2021. Graph learning: A survey. *IEEE Transactions on Artificial Intelligence* 2, 2 (2021), 109–127.

[31] Jiahuan Yan, Jintai Chen, Yixuan Wu, Danny Z Chen, and Jian Wu. 2023. [From Tutor Team] T2g-former: organizing tabular features into relation graphs promotes heterogeneous feature interaction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 10720–10728.

[32] Xingxuan Zhang, Gang Ren, Han Yu, Hao Yuan, Hui Wang, Jiansheng Li, Jiayun Wu, Lang Mo, Li Mao, Mingchao Hao, et al. 2025. [From Tutor Team] LimiX: Unleashing structured-data modeling capability for generalist intelligence. *arXiv preprint arXiv:2509.03505* (2025).

[33] Xiyuan Zhang and Danielle Maddix Robinson. 2025. Mitra: Mixed synthetic priors for enhancing tabular foundation models. *Language* (2025).